

Ex.No:11**Date:07/10/25**

EXAMINING NETWORK ADDRESS TRANSLATION (NAT) USING CISCO PACKET TRACER

Aim:

To Examine Network Address Translation (NAT) using Cisco Packet Tracer.

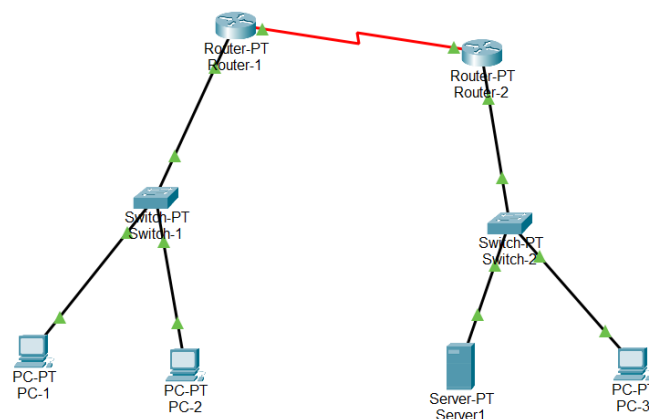
1. Setting Up the Network Topology

Devices Required:

- 2 Routers (R1, R2)
- 2 Switches (S1, S2)
- 3 PCs (PC1, PC2, PC3)
- 1 Server
- Serial and Ethernet cables

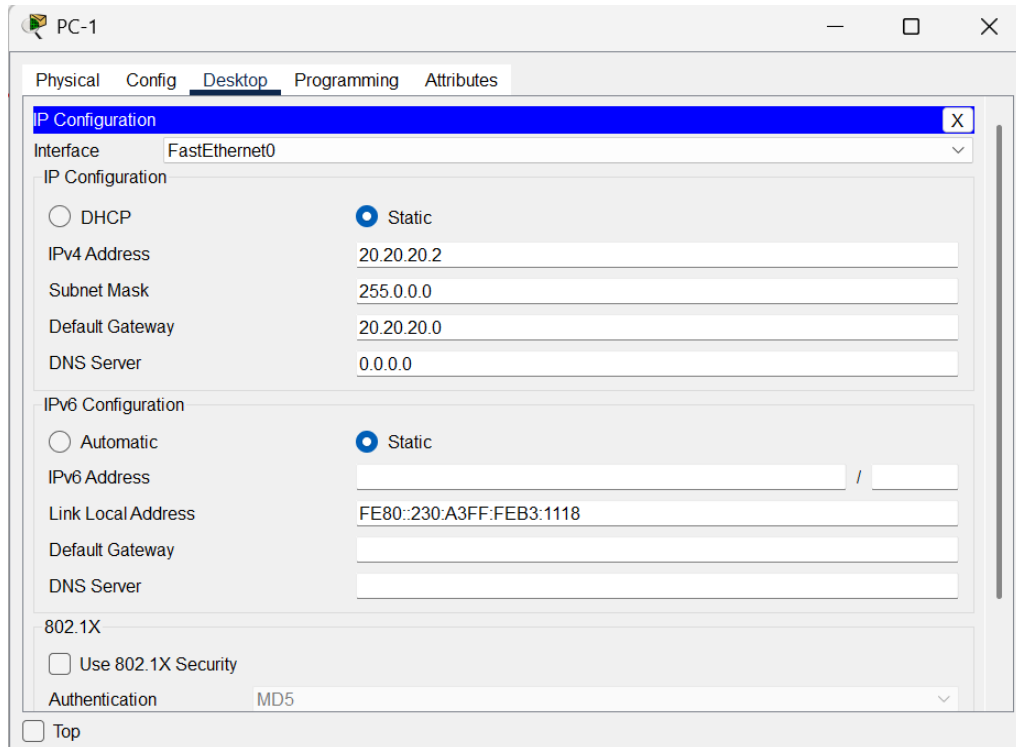
Cable the devices:

- PC1 → FastEthernet0 → Switch S1.
- PC2 → FastEthernet0 → Switch S1.
- Switch S1 → GigabitEthernet0/0 → Router R1.
- Server → FastEthernet0 → Switch S2.
- Switch S2 → GigabitEthernet0/0 → Router R2.
- Router R1 Serial0/0/0 ↔ Serial0/0/0 Router R2 (serial cable).



2.Assign IP Addresses to PC and Server:

1. Open the PC.
2. Go to Desktop → IP Configuration.
3. Enter the details:



PC-1

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 20.20.20.2

Subnet Mask: 255.0.0.0

Default Gateway: 20.20.20.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::230:A3FF:FE83:1118

Default Gateway:

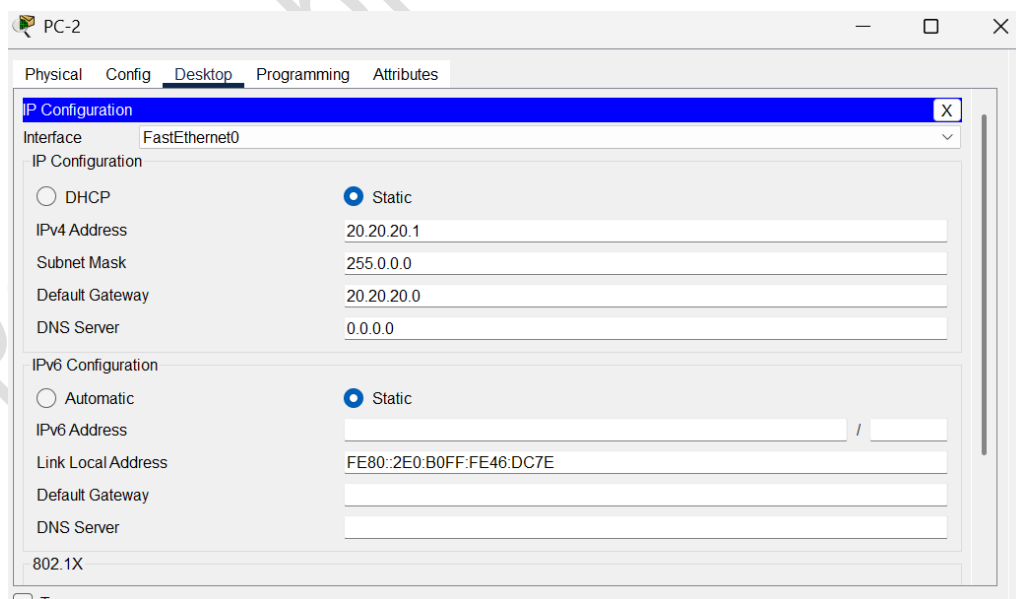
DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

☐ Top



PC-2

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 20.20.20.1

Subnet Mask: 255.0.0.0

Default Gateway: 20.20.20.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

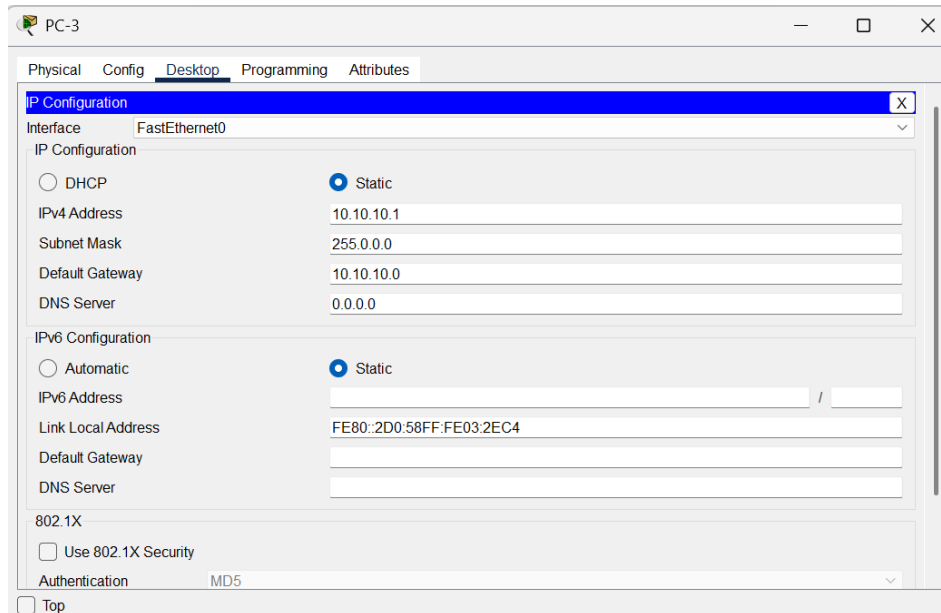
Link Local Address: FE80::2E0:B0FF:FE46:DC7E

Default Gateway:

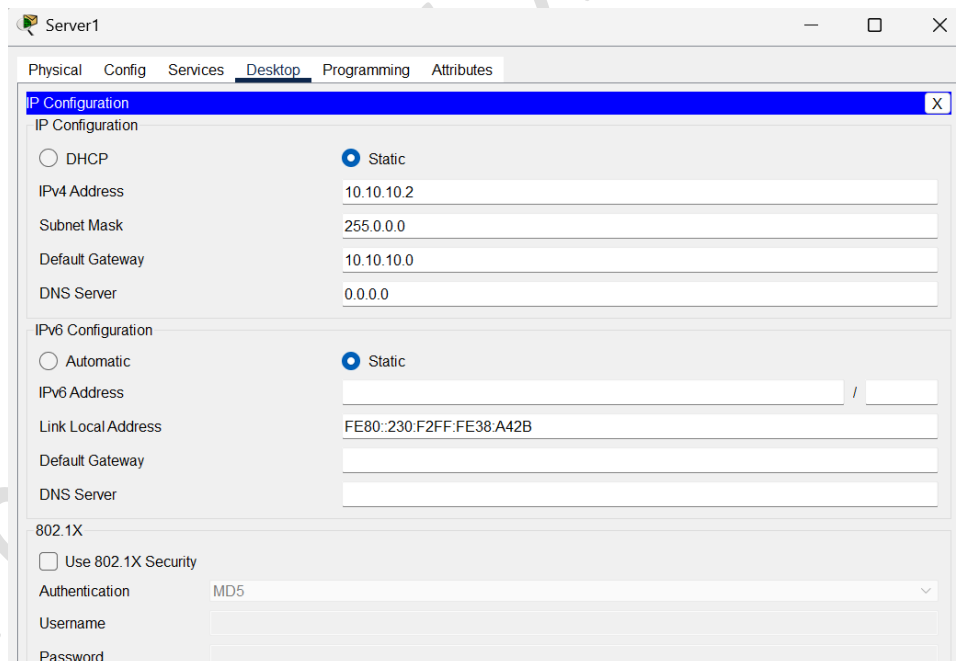
DNS Server:

802.1X

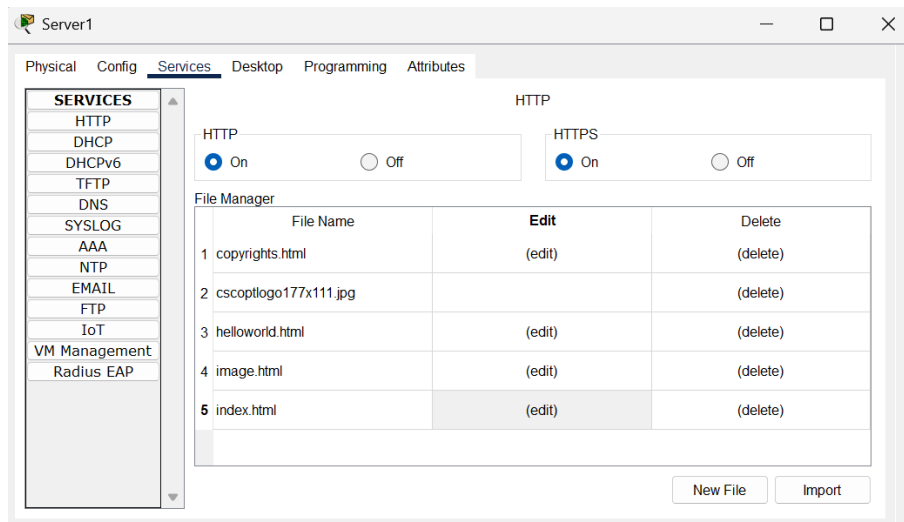
☐ Top



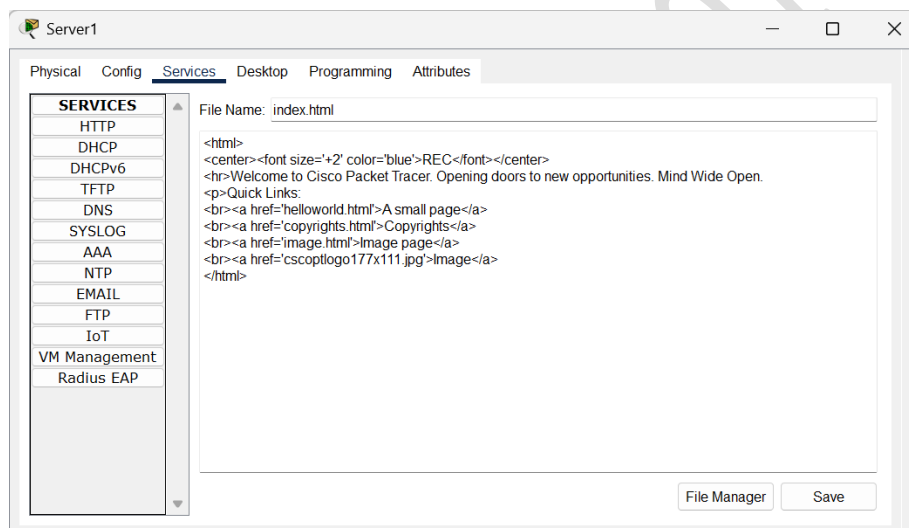
1. Open the Server.
2. Go to Desktop → IP Configuration.
3. Enter the details:



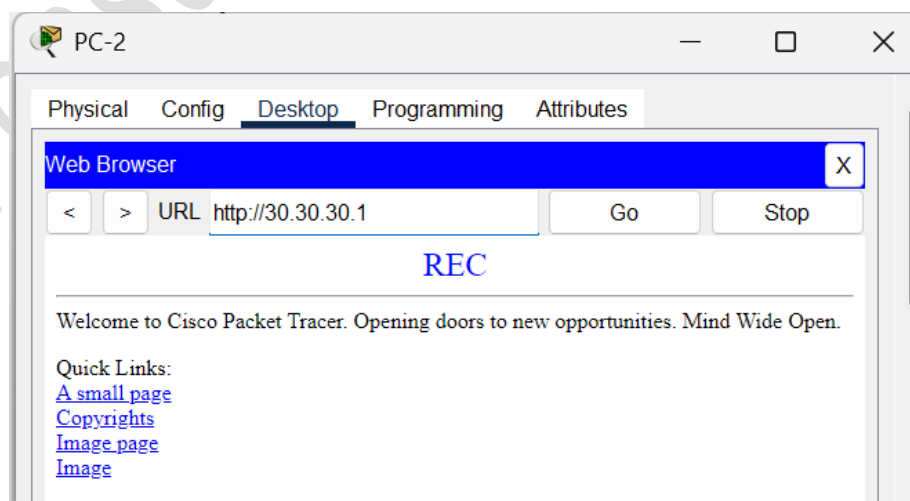
1.Go to services tab,edit index.html



2.Change the heading.



3.Open browser in PC1 and type the following url. Notice the Change in heading.



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3.Configure the routers

- Click on the router,go to config tab.
- Under interface,select the port connected . Mark the tick box “on”.
Fill the ip address and subnet mask.
- Repeat for all the ports in both the routers.

The screenshot shows the configuration window for Router-1, specifically the 'Config' tab for the 'FastEthernet0/0' interface. The left sidebar shows a tree view with 'INTERFACE' expanded and 'FastEthernet0/0' selected. The main area displays the following settings:

- Port Status:** On (checked)
- Bandwidth:** 100 Mbps (selected), 10 Mbps (unselected)
- Duplex:** Full Duplex (selected), Half Duplex (unselected)
- MAC Address:** 0009.7C77.CCA5
- IP Configuration:**
 - IPv4 Address: 20.20.20.0
 - Subnet Mask: 255.0.0.0
- Tx Ring Limit:** 10

Below the configuration fields, the 'Equivalent IOS Commands' section shows the following commands:

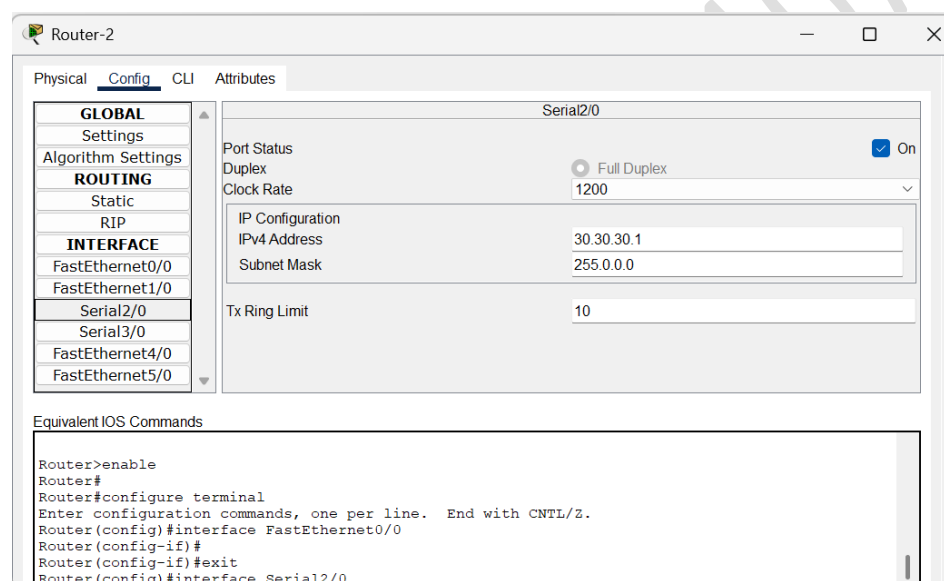
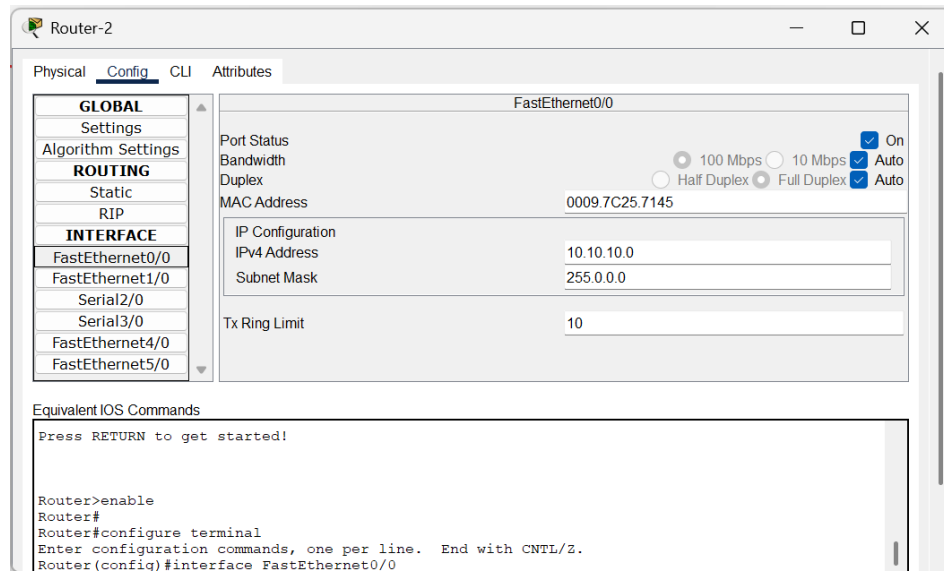
```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
```

The screenshot shows the configuration window for Router-1, specifically the 'Config' tab for the 'Serial2/0' interface. The left sidebar shows a tree view with 'INTERFACE' expanded and 'Serial2/0' selected. The main area displays the following settings:

- Port Status:** On (checked)
- Duplex:** Full Duplex (selected)
- Clock Rate:** 2000000
- IP Configuration:**
 - IPv4 Address: 30.30.30.2
 - Subnet Mask: 255.0.0.0
- Tx Ring Limit:** 10

Below the configuration fields, the 'Equivalent IOS Commands' section shows the following commands:

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#
```



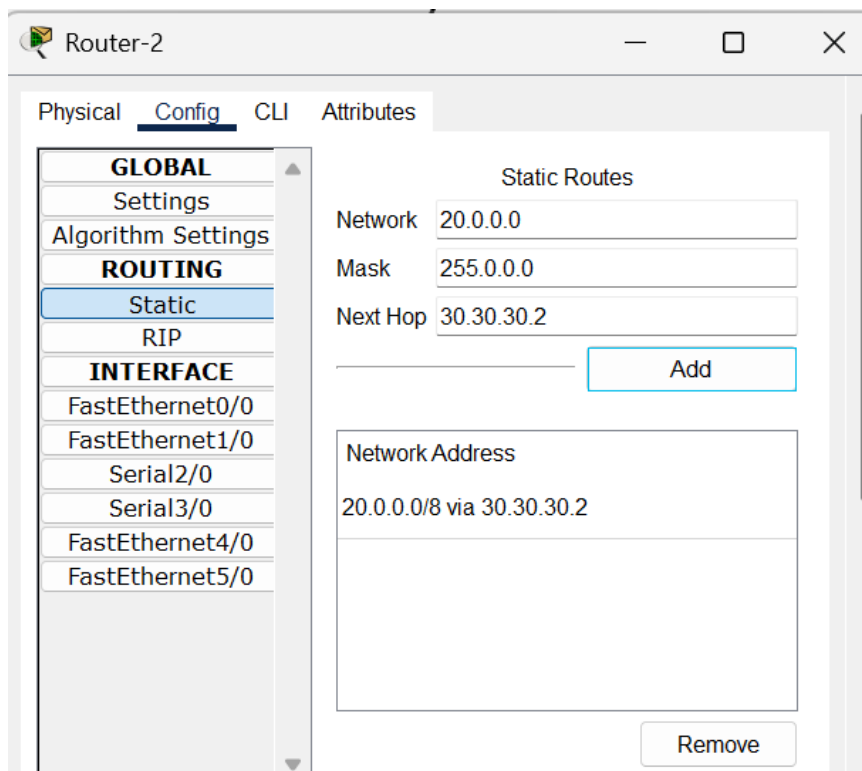
Static NAT Mapping on Router1:

- Open R1 → CLI
- Enter:

```
Router(config)#ip nat inside source static 10.10.10.1 30.30.30.1
Router(config)#ip nat inside source static 10.10.10.2 30.30.30.1
Router(config)#int f0/0
Router(config-if)#ip nat inside
Router(config-if)#int s2/0
Router(config-if)#ip nat outside
Router(config-if)#exit
```

Static Routing on Router2:

- Open Router2 → Config tab → Static.
- Fill the fields: Network, Subnet Mask, Next Hop
- Click Add



4. Test Connectivity Using PDU and Ping

PDU Test:

- Select the Add Simple PDU tool from the main toolbar.
- Source: PC3 → Destination: PC2.
- Send the PDU.
- The PDU succeeds because the private IP can traverse the public network.

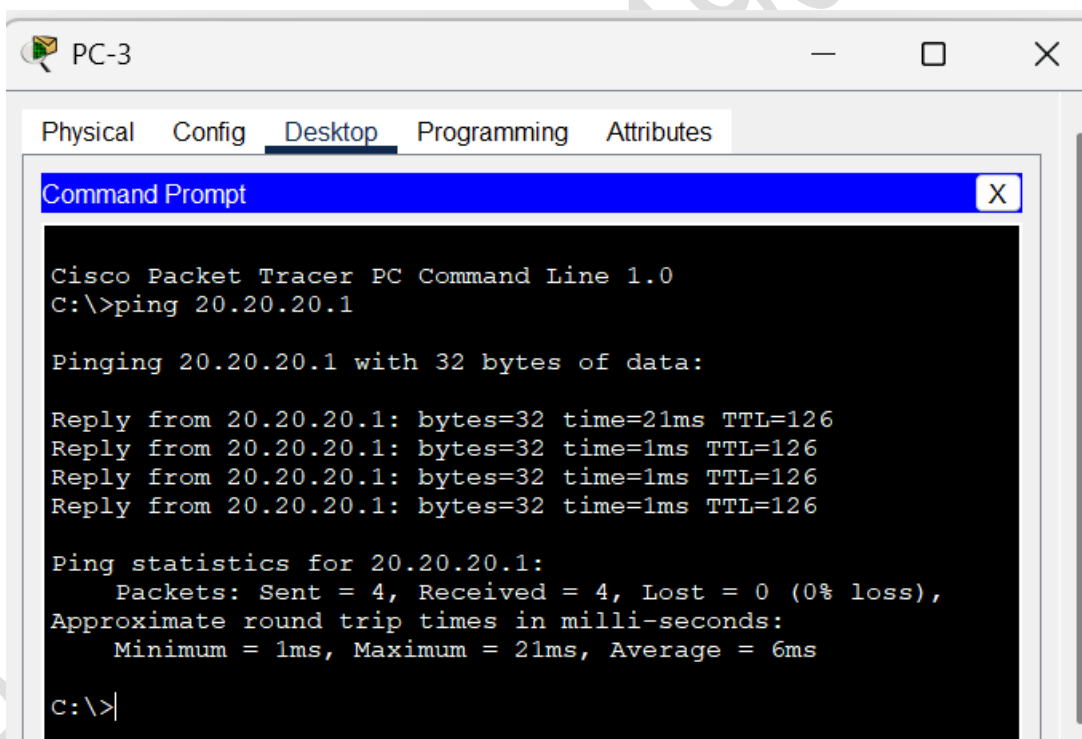
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC-3	PC-2	ICMP		0.000	N	0	(edit)	(delete)

- Select the Add Simple PDU tool from the main toolbar.
- Source: PC2 → Destination: PC3.
- Send the PDU.
- The PDU fails because the public IP cannot traverse the private network.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC-3	PC-2	ICMP		0.000	N	0	(edit)	(delete)
	Failed	PC-2	PC-3	ICMP		0.000	N	1	(edit)	(delete)

Ping test:

- Open command prompt in PC3 and ping PC2.
- Ping successful.



```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 20.20.20.1

Pinging 20.20.20.1 with 32 bytes of data:

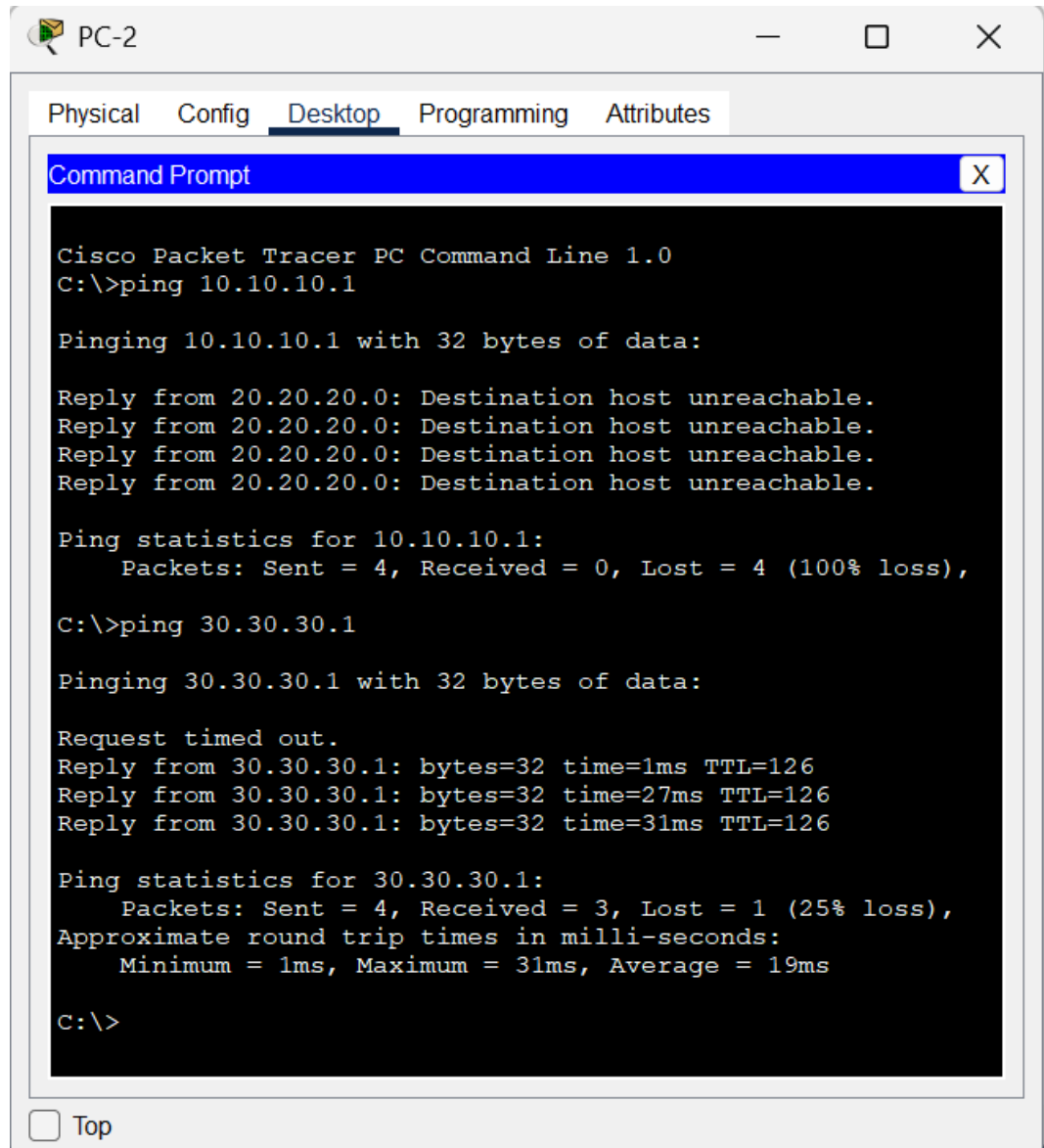
Reply from 20.20.20.1: bytes=32 time=21ms TTL=126
Reply from 20.20.20.1: bytes=32 time=1ms TTL=126
Reply from 20.20.20.1: bytes=32 time=1ms TTL=126
Reply from 20.20.20.1: bytes=32 time=1ms TTL=126

Ping statistics for 20.20.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 21ms, Average = 6ms

C:\>|
  
```

- Open command terminal in PC2 and ping PC3.

- Ping Failed.



The screenshot shows a Cisco Packet Tracer PC Command Prompt window for PC-2. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, showing a Command Prompt window. The Command Prompt displays the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:

Reply from 20.20.20.0: Destination host unreachable.
Reply from 20.20.20.0: Destination host unreachable.
Reply from 20.20.20.0: Destination host unreachable.
Reply from 20.20.20.0: Destination host unreachable.

Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 30.30.30.1

Pinging 30.30.30.1 with 32 bytes of data:

Request timed out.
Reply from 30.30.30.1: bytes=32 time=1ms TTL=126
Reply from 30.30.30.1: bytes=32 time=27ms TTL=126
Reply from 30.30.30.1: bytes=32 time=31ms TTL=126

Ping statistics for 30.30.30.1:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 31ms, Average = 19ms

C:\>
```

At the bottom of the Command Prompt window, there is a checkbox labeled "Top".

Result:

Thus Network Address Translation (NAT) was examined using Cisco Packet Tracer.